

Sprint 2 Retrospektive

Projekt: KI-gestütztes System zur frühzeitigen Erkennung von Datenlecks

Sprint: Sprint 2

The primary goal of Sprint 2 was to build a more functional, analysis-focused, and user-friendly system architecture on top of the technical foundation established in Sprint 1. During this sprint, the team focused specifically on data analysis, calculating risk scores, classifying findings, developing the alert mechanism, supporting the analyst workflow, and ensuring the frontend dashboard worked in tandem with the backend. Additionally, integrating new data sources into the system and ensuring these sources could be managed more effectively was a key objective of the sprint. As a result, Sprint 2 marked a significant milestone in the project's evolution from a mere data-collecting prototype to a system capable of evaluating data and making it more meaningful for analysts.

Throughout the sprint, team members worked in parallel on different system components. On the backend side, we focused specifically on developing alert generation and analyst workflows. In this context, we worked on determining at which risk levels alerts should be generated, how analysts could review findings, and how filtering and search processes would be supported. However, before proceeding with direct endpoint development in this area, a technical issue arose on the database side. Because multiple heads had formed in the Alembic migration chain, the structure of the database changes became inconsistent. This situation was assessed as a problem that needed to be resolved first to ensure the backend development could continue safely. Therefore, the team first corrected the migration structure and stabilized the database side. As a result, backend development could continue on a more solid foundation. By the end of the sprint, the backend was largely complete; the next phase will focus on integration improvements, final adjustments, and addressing any future bugs that may arise.

The goal of the Analysis Engine was to detect signs of data leakage in documents received from the collector and parser, calculate a risk score, and classify findings into meaningful categories. To this end, the system was developed to search for risky patterns such as email-password combinations, hash values, database dump statements, API keys, private keys, and secret/config information. Additionally, company profiles were checked for matches in company names, aliases, and domains. This ensured that the system would consider not only publicly known company names but also alternative names or domain information associated with the company.

In this section, using regex-based pattern detection was a sound technical decision, as leaked data often follows specific formats. In addition, terms specific to leak terminology—such as “combo list,” “database dump,” “leaked,” and “credentials”—were also incorporated into the analysis process. One of the most significant challenges encountered during the Sprint was the false positive problem. Since spam content, repetitive information, and words that can have different meanings depending on context are frequently found in darknet and paste data, relying on a single word or a single pattern to make a decision would not be sufficiently reliable. Therefore, in Sprint 2, a key decision was made to evaluate multiple signals together. It was

concluded that when company matching, domain information, credential patterns, and leak terminology are present together, the risk score would be more reliable and explainable.

On the frontend side, the dashboard was largely designed and integrated with the backend APIs. During the sprint, the frontend layout was revisited several times and significantly revised about three or four times. Although these changes initially required additional time, they helped make the dashboard more suitable for analysts' needs and easier to understand. Findings, alerts, visualizations, and basic page structures have been added to the frontend. As a result, the frontend has evolved from being merely a visual mockup to a structure capable of working with backend data. However, minor bugs, design improvements, and some integration adjustments are still ongoing.

Work was conducted on identifying new source links and adding them to the system. This process proved more challenging than expected, as it required the careful selection of new sources that were both available and relevant to the project. A new configuration file was created to help organize the incoming links more effectively. This decision will make it easier to manage and verify new sources in the future and ensure they are incorporated into the collector process in a more organized manner.

During Sprint 2, the team encountered several technical and operational challenges. On the backend side, inconsistencies in the database migration structure required additional work during part of the development process. As a result, the existing database structure had to be reorganized and stabilized before moving on to new features. On the Analysis Engine side, ensuring that the data could be interpreted reliably became a key priority. In particular, due to the possibility that certain content might be incorrectly flagged as risky, the analysis logic was reviewed, and it was decided that evaluations would be based on multiple criteria. On the frontend side, the dashboard design was revisited several times throughout the development process. Although this initially required additional time, it resulted in a more intuitive and user-friendly interface. Additionally, identifying new data sources and organizing them appropriately for the system required more research and organizational work than anticipated. Consequently, a more structured framework was established to streamline source management, enabling centralized control over these sources.

To ensure quality during the sprint, the different system components were checked separately. On the backend side, the corrected database migration structure and the basic functionality of the implemented endpoints were reviewed. In the Analysis Engine, the detection of different risky patterns, company matching logic, and risk score calculation were tested with example findings. On the frontend side, the dashboard rendering, findings and alerts views, filtering options, and visualizations were checked. However, more comprehensive integration tests with realistic data will be necessary in the next sprint, especially to verify the interaction between backend, Analysis Engine, and frontend.

During the meeting with the instructor, it was decided that the number of findings should be increased and that the dashboard should display data not only on a weekly basis but also on a monthly basis. This feedback is important because it allows the dashboard to show not only short-term developments but also longer-term trends. This will enable the analyst to evaluate findings in the system over broader time frames and track developments more clearly.

The primary focus for the next sprint will be on better integrating existing components and resolving remaining minor bugs. Since major backend developments are largely complete, the

focus will now shift to integration improvements, final adjustments, and fixing any emerging issues. Another key next step is to display Analysis Engine results on the dashboard in a clearer and more understandable manner. In particular, the identified companies, detected patterns, risk score, and classification description should be easily interpretable for the analyst. On the frontend side, final design adjustments to the dashboard should be made, the number of findings should be increased, and monthly data visualization should be added alongside the weekly view.

Overall, Sprint 2 can be considered successful, as the project made an important step from a basic prototype toward a more functional and analysis-oriented system. The system is now closer to not only collecting data, but also evaluating it, assigning risk scores, generating alerts, and presenting the results in a way that is useful for analysts. The technical issues encountered during the sprint, especially the database migration problem and the false positive risk in the Analysis Engine, were identified early and handled in a structured way. These experiences also helped define clearer improvement areas for the next sprint.